

## ALTURAS, CA, USA

WMO: 725958

Lat: 41.483N

Lon: 120.567W

Elev: 1334

StdP: 86.30

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 94299

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 12            | -15.8      | -12.1 | -18.4                         | 0.9 | -11.5 | -15.2 | 1.2 | -7.2 | 11.5                  | 5.3  | 9.8 | 4.1  | 0.4                   | 20   | 0.601 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 22.8                   | 34.7            | 15.6 | 33.0 | 15.1 | 31.6 | 14.5 | 16.9                | 30.8 | 16.0 | 29.8 | 15.3 | 29.1 | 4.2                  | 270  |

| Dehumidification DP/MCDB and HR |      |      |      |     |      |     |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|-----|------|-----|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |     |      | 2%  |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR  | MCDB | DP  | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e) | (f)  | (g) | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 12.3                            | 10.5 | 19.3 | 10.8 | 9.4 | 19.1 | 9.1 | 8.5 | 18.1 | 52.2          | 30.8 | 49.2 | 29.8 | 46.8 | 29.0 | 20.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 9.9           | 8.5  | 7.6 | -20.9                      | 37.5 | 4.5                | 1.4 | -24.2                                              | 38.5 | -26.8      | 39.4 | -29.4      | 40.2 | -32.6      | 41.2 |    |    |
| (5)               |      |     | -21.1                      | 18.6 | 4.5                | 1.2 | -24.3                                              | 19.4 | -26.9      | 20.1 | -29.4      | 20.8 | -32.6      | 21.7 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 8.7         | -0.1  | 1.2   | 4.0   | 6.6   | 11.3  | 15.9  | 20.2  | 18.9  | 14.6  | 8.5   | 3.2   | -0.7  |  |  |
|                                                                        | DBStd       | 8.09        | 4.72  | 3.90  | 3.58  | 3.78  | 3.87  | 3.74  | 2.80  | 2.63  | 3.67  | 3.66  | 4.37  | 4.55  |  |  |
|                                                                        | HDD10.0     | 1505        | 314   | 246   | 187   | 112   | 32    | 4     | 0     | 0     | 7     | 70    | 204   | 331   |  |  |
|                                                                        | HDD18.3     | 3653        | 572   | 479   | 444   | 351   | 220   | 88    | 13    | 24    | 117   | 304   | 453   | 589   |  |  |
|                                                                        | CDD10.0     | 1030        | 0     | 0     | 1     | 11    | 71    | 181   | 317   | 276   | 146   | 25    | 2     | 0     |  |  |
|                                                                        | CDD18.3     | 136         | 0     | 0     | 0     | 0     | 1     | 16    | 72    | 42    | 7     | 0     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 5033        | 0     | 0     | 1     | 13    | 160   | 716   | 1941  | 1494  | 655   | 54    | 0     | 0     |  |  |
|                                                                        | CDH26.7     | 2256        | 0     | 0     | 0     | 1     | 37    | 269   | 998   | 697   | 243   | 11    | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 2.4         | 1.9   | 2.4   | 2.9   | 3.0   | 2.9   | 2.8   | 2.5   | 2.3   | 2.0   | 2.0   | 2.0   | 2.1   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 359         | 40    | 33    | 41    | 41    | 44    | 22    | 5     | 8     | 9     | 25    | 41    | 49    |  |  |
|                                                                        | PrecMax     | 598         | 98    | 79    | 106   | 98    | 141   | 61    | 21    | 30    | 36    | 78    | 107   | 135   |  |  |
|                                                                        | PrecMin     | 210         | 2     | 9     | 14    | 9     | 0     | 3     | 0     | 0     | 0     | 0     | 6     | 12    |  |  |
|                                                                        | PrecStd     | 111         | 29    | 18    | 23    | 20    | 35    | 19    | 5     | 9     | 10    | 21    | 24    | 39    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 14.3  | 16.3  | 21.3  | 25.2  | 30.5  | 34.7  | 37.4  | 35.9  | 33.6  | 28.5  | 20.8  | 12.2  |  |  |
|                                                                        |             | MCWB        | 6.2   | 7.5   | 9.3   | 11.3  | 14.4  | 15.9  | 17.2  | 16.1  | 14.9  | 12.5  | 9.8   | 6.0   |  |  |
|                                                                        | 2%          | DB          | 11.7  | 13.0  | 17.9  | 22.4  | 27.4  | 32.0  | 35.2  | 33.9  | 31.3  | 24.2  | 17.6  | 10.0  |  |  |
|                                                                        |             | MCWB        | 5.5   | 5.6   | 7.9   | 10.2  | 13.3  | 15.0  | 15.7  | 15.3  | 13.6  | 10.9  | 8.6   | 5.0   |  |  |
|                                                                        | 5%          | DB          | 9.1   | 10.6  | 15.5  | 19.6  | 25.0  | 29.8  | 33.7  | 32.4  | 29.5  | 22.4  | 14.1  | 8.1   |  |  |
|                                                                        |             | MCWB        | 4.5   | 5.2   | 6.9   | 9.0   | 12.0  | 13.8  | 15.3  | 14.7  | 13.0  | 10.0  | 6.9   | 4.5   |  |  |
|                                                                        | 10%         | DB          | 7.1   | 8.2   | 12.4  | 16.9  | 22.4  | 27.7  | 32.3  | 30.9  | 27.6  | 19.6  | 11.3  | 6.4   |  |  |
|                                                                        |             | MCWB        | 3.7   | 3.9   | 5.8   | 7.7   | 11.0  | 13.3  | 15.0  | 14.1  | 12.7  | 9.0   | 5.9   | 3.6   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 8.4   | 8.5   | 10.1  | 12.4  | 15.5  | 17.1  | 18.5  | 18.3  | 16.6  | 13.1  | 10.7  | 7.5   |  |  |
|                                                                        |             | MCDB        | 12.6  | 14.4  | 19.1  | 22.5  | 27.7  | 31.3  | 33.6  | 33.0  | 27.6  | 24.1  | 18.6  | 9.9   |  |  |
|                                                                        | 2%          | WB          | 6.4   | 6.9   | 8.7   | 10.8  | 13.9  | 15.6  | 17.3  | 16.5  | 15.1  | 11.7  | 9.1   | 6.2   |  |  |
|                                                                        |             | MCDB        | 10.0  | 11.6  | 16.2  | 20.8  | 25.5  | 29.4  | 31.7  | 30.1  | 27.6  | 22.2  | 16.0  | 8.7   |  |  |
|                                                                        | 5%          | WB          | 5.0   | 5.4   | 7.5   | 9.5   | 12.7  | 14.7  | 16.3  | 15.5  | 13.9  | 10.6  | 7.8   | 5.1   |  |  |
|                                                                        |             | MCDB        | 8.4   | 9.5   | 14.3  | 18.3  | 23.0  | 27.8  | 30.5  | 29.3  | 27.2  | 20.7  | 13.4  | 7.4   |  |  |
|                                                                        | 10%         | WB          | 3.8   | 4.2   | 6.3   | 8.2   | 11.5  | 13.7  | 15.5  | 14.7  | 12.9  | 9.5   | 6.4   | 3.8   |  |  |
|                                                                        |             | MCDB        | 6.8   | 8.1   | 11.9  | 15.7  | 20.7  | 26.0  | 29.9  | 28.6  | 26.5  | 18.3  | 10.7  | 6.2   |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 12.4  | 12.4  | 14.0  | 15.0  | 16.7  | 19.7  | 22.8  | 22.9  | 22.4  | 19.2  | 14.3  | 11.5  |  |  |
|                                                                        |             | MCDBR       | 16.5  | 16.7  | 21.0  | 22.6  | 22.8  | 24.1  | 24.7  | 25.4  | 26.5  | 25.4  | 20.5  | 14.4  |  |  |
|                                                                        | 5% WB       | MCWBR       | 10.8  | 10.3  | 11.7  | 11.5  | 10.1  | 9.3   | 9.2   | 10.4  | 12.0  | 13.3  | 12.3  | 9.9   |  |  |
|                                                                        |             | MCWBR       | 13.5  | 14.2  | 18.4  | 20.6  | 20.5  | 22.4  | 22.8  | 23.0  | 24.2  | 22.3  | 18.0  | 10.7  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.279 | 0.289 | 0.300 | 0.307 | 0.314 | 0.308 | 0.312 | 0.310 | 0.298 | 0.297 | 0.280 | 0.275 |  |  |
|                                                                        |             | taud        | 2.507 | 2.510 | 2.505 | 2.493 | 2.513 | 2.557 | 2.545 | 2.530 | 2.560 | 2.551 | 2.563 | 2.531 |  |  |
|                                                                        | Ebn at Noon | Ebn at Noon | 887   | 931   | 956   | 966   | 961   | 963   | 955   | 950   | 945   | 909   | 882   | 863   |  |  |
|                                                                        |             | Edh at Noon | 72    | 84    | 95    | 103   | 104   | 100   | 100   | 98    | 89    | 79    | 68    | 65    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 1.98  | 2.88  | 4.06  | 5.43  | 6.72  | 7.88  | 8.00  | 7.07  | 5.62  | 3.77  | 2.23  | 1.58  |  |  |
|                                                                        | RadStd      | RadStd      | 0.26  | 0.35  | 0.39  | 0.53  | 0.60  | 0.41  | 0.34  | 0.33  | 0.28  | 0.32  | 0.23  | 0.31  |  |  |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only           | N/A   | N/A     | N/A    | +0.59   | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (7 neighbors) | N/A   | N/A     | N/A    | +0.60   | N/A   | N/A   | N/A         | N/A     | +91     | +42     |

Nomenclature: See separate page